

Supplementary Material: Learning Joint Recovery of Multi-Agent Graph Optimization for Satellite–Ground Scheduling

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1 Feature and Execution Details

Observable inputs. The reward scale is $s_G = \max\{1, \sum_i w_i / |V|\}$. Each recovery vertex supplies normalized reward, log degree, local degree fraction, actual warm membership, log factor memberships and log maximum incident factor size. The structural encoder has two layers of width 32. Its shared mean and reward-weighted pooled encodings each have width 32. The seven request features are the seconds/nodes/iterations indicators, log native allowance, log remaining seconds, normalized actual warm value and actual warm fraction. The thirteen summary features are total, maximum, mean and standard deviation of normalized recovery reward; log edge count and density; log factor count, maximum/mean factor size, mean memberships and high-order factor fraction; clique-load priority P1; and the clique-partition upper value. The immediate gain is an exact offset, not a learned input. The evaluated version uses these features directly; it does not contain the subsequent untrained normalization proposal.

Cache boundary. A structural unit is identified by its graph/model version, ordered recovery vertices, factors, actual warm membership and reward scale. Different native allowances share a unit. Changed warm membership creates a new unit. No target membership, execution score or previous graph enters the cache. Offline training uses attached encodings; online encodings are detached. Each decision starts with empty caches. Packing, bound calculation, device transfer, cache lookup and inference are charged to the decision.

Fractional capacity property. Let $s_F = \sum_{j \in F} p_j$ and define

$$d_i = \max\{1, \max_{F \ni i} s_F\}.$$

For each factor F with $s_F > 0$, $d_i \geq s_F$ for every $i \in F$, giving $\sum_{i \in F} p_i / d_i \leq s_F / s_F = 1$. When $s_F = 0$ the sum is zero. Thus simultaneous contraction satisfies the covered clique inequalities. It does not establish integral feasibility, and the fractional weighted sum is only an input to the bounded scalar head. Feasible memberships are produced and checked by the executor.

Table 1: All 15 completed fits. Regret ↓ is the selected conditional allocation proxy, in fractions of original incumbent value. It is not complete online return quality.

Variant	Seed	Epoch	Regret ↓
Capacity	17	1	0.204938
Capacity	29	1	0.204938
Capacity	43	1	0.204938
FreeOccupancy	17	5	0.204060
FreeOccupancy	29	1	0.204938
FreeOccupancy	43	19	0.203691
NoAux	17	1	0.204938
NoAux	29	12	0.201069
NoAux	43	14	0.195201
NoWarm	17	1	0.204938
NoWarm	29	1	0.204938
NoWarm	43	1	0.204938
CheapSummary	17	1	0.204938
CheapSummary	29	1	0.204938
CheapSummary	43	1	0.204938

Bounds and empty cases. The warm lower value is from an actually executed feasible membership. The upper value sums the maximum reward in each disjoint clique block plus leftover singleton rewards. Hence every independent recovery lies below it. Empty recovery scopes have zero recovery; empty loss means are zero. Floating summation noise may raise an upper value to its warm lower value within the declared absolute/relative tolerance, but cannot replace original integer objectives or membership checks.

2 Complete Fitting Configuration

The fixed pool contains 72 training and 24 development validation graphs, split by graph before snapshot and alternative extraction. All five variants use seeds 17, 29 and 43, Adam learning rate 10^{-3} , four-graph batches and 40 epochs. The first minimum conditional allocation regret selects a checkpoint. Table 1 gives every selection; repeated epochs or fit seeds are not additional independent graphs.

Eleven histories have exactly constant conditional regret although parameters and auxiliary losses change. The unchanged proxy does not imply that gradients through the entire model disappear. The main paper’s online comparison uses all selected fits rather than choosing a favorable fitting seed.

3 Benchmark Roles and Cost Boundaries

The main controlled comparison uses all 24 validation graphs, eleven policies and three total deadlines. Each learned policy has three fitting seeds, giving 1,512 runs. Within each graph, fit means precede equal menu/resource aggregation. Zero opportunities, failures and late returns retain their weight. The fixed common greedy prefix is included in total returned quality and subtracted separately for additional recovery. A result received after its deadline has zero strict gain regardless of an internally available candidate.



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The public profile retains all 40 original unit-weight graphs: 16 DIMACS clique complements, 18 SATLIB UF and six CBS conflict graphs. Seven native configurations at two soft workpoints give 1,680 runs; three released neural recipes give 360, for 2,040 runs total. Family means have equal mass after three initial schedules are averaged within each graph. Complete native cold cost includes export, startup, preprocessing, parsing and validation. Resident neural request cost excludes once-only setup. DIFUSCO one/four-sample parent/worker setup is 5.948/5.785 and 5.946/5.828 seconds; COExpander setup is 7.238/7.125 seconds. These boundaries preclude a common-budget speedup claim.

Whole-graph profiles compare published solvers and establish context for the shared repair kernel. They do not test our learned policy on the same public graphs. Released checkpoint overlap remains unresolved. The current controlled development results also do not establish independent confirmation or full application-scale transfer. The companion data tables retain these roles explicitly; future experimental designs are not presented as completed evidence.

4 Research Assistance Disclosure

OpenAI Codex, using GPT-6-family agents, assisted with method formulation, experiment design, implementation, analysis, drafting and deterministic scientific plotting. Human authors remain responsible for the research claims, references and final submission. The retained instructions included:

“The value of a coordinated adjustment depends on both its immediate gain and the mutually compatible contact opportunities that can be recovered afterward.”

“Training compares feasible outcomes obtained by applying alternative actions to the same state under a shared, budgeted execution procedure.”

The authors requested formally published baselines, public graph tasks, complete-cost comparisons, separate section agents, compact consistent figures, and separation of operational reports from the paper. Their final instruction froze the completed version rather than continuing exploratory search. Exact identifiers and complete prompts for every earlier model run were not preserved; this disclosure does not reconstruct missing records. Figures are deterministic vector diagrams or plots of recorded results. Illustrative graph weights are mathematical examples, not experimental outputs.